Online Resource 8: PRISMA 2020 checklist

Article: Explainable Deep Learning and Class-Imbalance Mitigation in Network Intrusion Detection: A Structured Review of Methods, Evaluation Practice and Reporting Gaps. **Journal:** Archives of Computational Methods in Engineering. **Authors:** Tarun Jangwal, Purva Pathania (Lovely Professional University, Phagwara, Punjab, India). **Corresponding author:** Tarun Jangwal, tarun.24044@lpu.co.in.

Item wording is abbreviated from the PRISMA 2020 checklist (Page MJ, McKenzie JE, Bossuyt PM, et al. The PRISMA 2020 statement. BMJ 2021;372:n71; CC BY 4.0). "Location" refers to the revised manuscript. Items marked as not reported or not performed are declared in the manuscript and are not reconstructed.

Table 1 PRISMA 2020 checklist for the review

Section	Item	Checklist item (abbreviated)	Location in manuscript	Status
Title	1	Identify the report as a systematic review	Title	Not applicable: reported as a structured review because identification is not reproducible; PRISMA 2020 used as reporting format (Section 2.1)
Abstract	2	Structured summary (PRISMA 2020 for Abstracts)	Abstract	Partly reported: registration and funding are not stated in the abstract (no registration exists; funding is an author action)
Introduction	3	Rationale in the context of existing knowledge	Sections 1.1, 1.2, 5	Reported
Introduction	4	Objectives or questions	Section 1.3, Table 1	Reported
Methods	5	Eligibility criteria and grouping for synthesis	Section 2.4, Table 3; Sections 2.8–2.10	Reported
Methods	6	Information sources and date last searched	Section 2.2; Fig. 2	Not reported for identification: sources and dates were not recorded (Section 9.1). A dated post-hoc OpenAlex coverage query (3 November 2026) is reported in Section 2.2 but was not used to identify studies
Methods	7	Full search strategies	Section 2.2, Table 2	Partly reported: keyword families only (Table 2); exact strings were not recorded. The full Boolean string of the post-hoc coverage query is given in Section 2.2
Methods	8	Selection process (reviewers, independence, automation tools)	Sections 2.5, 2.6, 2.7; final declaration (tools)	Reported (single pass by one reviewer, tools declared; no second reviewer)
Methods	9	Data collection process	Section 2.8; final declaration (tools)	Partly reported: provenance of the structured extractions is not recorded
Methods	10a	Outcomes for which data were sought	Sections 2.8, 2.9	Reported (practice-level data items; no effect outcomes)
Methods	10b	Other variables for which data were sought	Section 2.8; Online Resource 1	Reported
Methods	11	Risk-of-bias assessment methods	Section 2.12	Not performed; reason stated
Methods	12	Effect measures	Section 2.11	Not applicable: no effect measures synthesised; counts and proportions reported
Methods	13a	Deciding which studies were eligible for each synthesis	Sections 2.8, 2.10; denominators in Section 4	Reported
Methods	13b	Data preparation for synthesis (missing data, conversions)	Section 2.10	Reported (conflict rules; missing values not imputed)
Methods	13c	Tabulation and visual display methods	Section 4 (Tables 6–18, Figs. 3–8)	Reported
Methods	13d	Synthesis methods and rationale	Section 2.11	Reported (structured narrative synthesis; SWiM)
Methods	13e	Methods to explore heterogeneity	Section 2.11	Not applicable (no meta-

Section	Item	Checklist item (abbreviated)	Location in manuscript	Status
				analysis)
Methods	13f	Sensitivity analyses	Sections 4.3, 4.5	Partly reported: bounds under resolution of extraction conflicts
Methods	14	Reporting-bias assessment methods	Section 9.1	Not performed; selection bias of the candidate pool discussed
Methods	15	Certainty assessment methods	Section 2.12	Not performed; reason stated
Results	16a	Results of the search and selection, with flow diagram	Section 4.1, Fig. 2	Reported (identification-stage counts not recorded)
Results	16b	Studies that appeared to meet criteria but were excluded, with reasons	Sections 2.6, 2.7; Table 4	Reported
Results	17	Characteristics of included studies	Tables 6–8; Online Resource 1	Reported
Results	18	Risk of bias in studies	–	Not performed
Results	19	Results of individual studies	Online Resource 1; Tables 7–15	Reported as classifications; effect estimates not tabulated because protocols are not comparable (Section 6.3)
Results	20a	Characteristics and risk of bias of studies contributing to each synthesis	Scope and denominator blocks of Sections 4.2–4.6	Partly reported (no risk of bias)
Results	20b	Results of all syntheses	Sections 4.2–4.7	Reported
Results	20c	Investigations of heterogeneity	–	Not applicable
Results	20d	Sensitivity analyses	Sections 4.3, 4.5	Partly reported (conflict bounds)
Results	21	Risk of bias due to missing results	–	Not assessed
Results	22	Certainty of evidence	–	Not assessed
Discussion	23a	Interpretation in the context of other evidence	Sections 5, 6	Reported
Discussion	23b	Limitations of the evidence	Section 9.2	Reported
Discussion	23c	Limitations of the review processes	Section 9.1	Reported
Discussion	23d	Implications for practice, policy and research	Sections 6.4, 7, 8	Reported
Other information	24a	Registration	Section 2.1	Not registered (stated)
Other information	24b	Protocol access	Section 2.1	No protocol (stated)
Other information	24c	Amendments to registration or protocol	Sections 2.6, 2.7, 2.9	Changes made during revision are described
Other information	25	Support (funding)	Statements and Declarations	Author action required
Other information	26	Competing interests	Statements and Declarations	Author action required
Other information	27	Availability of data, code and other materials	Data availability; Online Resources 1–7	Reported